

through the workflow matters as much as the result. Testing agentic systems requires a different discipline entirely, and most teams underinvest here until after a production incident forces the issue.

The emerging practice combines golden path replay (recording known-good trajectories and detecting deviation), adversarial probing (deliberate prompt injections and edge-case inputs), and sandboxed production simulation where agents operate against mirrored environments with real data shapes but no write access to live systems.

SLA design and SLO accountability: If an agentic workflow is in the critical path of a business process, it needs formal service level objectives. Latency budgets, availability targets, and degradation strategies — including fallback to a simpler model, a human-routed path, or a cached result — should be designed before go-live, not discovered during the first production incident.

Versioning and change management: When an agent's system prompt changes, its behaviour changes. When a tool

schema is updated, every agent that calls it is affected. When a new model version is deployed, behavioural regressions are possible without any code change at all. This is one of the more counterintuitive realities of agentic systems. Enterprise-grade agent management requires version-controlled agent definitions, semantic versioning for tool contracts, and the ability to roll back a deployed agent independently of the application it serves.

What frameworks give you and what they don't

The three pillars above describe what enterprise-ready agentic infrastructure requires. It's worth being equally precise about what the leading frameworks really provide, because the gap between the two is exactly where most production failures happen.

The popular frameworks each solve a real problem, and they solve it well. LangGraph gives you graph-based orchestration: explicit state machines where every node, edge, and transition is defined in code. It has built-in checkpointing backed by SQLite, Redis, or Postgres, and first-class human-in-the-loop support via interrupt nodes. It's running in production at LinkedIn and Uber, with LangSmith providing a solid observability layer on top. CrewAI gives you role-based multi-agent composition: agents as crew members with defined goals and tools, wired together rapidly with minimal boilerplate. It's the fastest path from idea to working prototype, claiming deployment across 60% of Fortune 500 companies as of its v1.0 GA in October 2025. Microsoft Agent Framework, the unified successor to AutoGen and Semantic Kernel, with GA targeted for end of Q1 2026, gives you graph-based workflows, checkpointing, and HITL on deep Azure integration, with native enterprise identity via Entra Agent ID.

A fourth category has emerged alongside these orchestration frameworks: platforms that approach the problem from the infrastructure layer up. Solace Agent Mesh, which reached general availability in December 2025, is the most distinct example. Built on Solace's event broker -- technology that enterprises like Bosch, RBC Capital Markets, and Heineken already run in mission-critical operations -- it treats agent communication as an event-driven messaging problem rather than an orchestration code problem. Every agent interaction flows through the event mesh. That means observability, fault tolerance, horizontal scaling, and message delivery guarantees are structural properties of the platform, not layers you add later. Security and governance sit in the gateway layer: SSO, RBAC, action-level permissions, and user-delegated access come as first-class capabilities. Gartner's August 2025 Innovation Insight on AI Agent Development Frameworks listed Solace as one of eight representative vendors in this space.

All of these are genuine capabilities, not marketing. But a precise reading of each reveals where the framework layer ends and the infrastructure layer begins, and which approach leaves you to assemble that infrastructure yourself

OPERATIONAL INTEGRITY
☐ Distributed trace ID propagated across all agent layers
☐ Decision provenance log persisted and queryable
☐ Compensating actions defined for every write operation
☐ Durable state store with explicit checkpoint markers
☐ Circuit breakers on all tool integrations
☐ HITL escalation path for confidence threshold breaches
SECURITY & TRUST
☐ Agent identity scoped to task, not user role
☐ Short-lived credentials; no persistent tokens in context
☐ Hard deny list with no escalation path
☐ Agent-to-agent delegation validated independently
☐ Data classification at each integration boundary
☐ PII/PHI handling mapped to compliance framework
GOVERNANCE AT SCALE
☐ Per-workflow budget cap and kill switch in place
☐ Cost attributed to team, workflow, and task
☐ Golden path regression test suite running in CI
☐ Adversarial probe scenarios documented and scheduled
☐ SLOs defined with explicit degradation strategy
☐ Agent definitions versioned; rollback procedure tested

Table 6: Enterprise readiness checklist